Ground-Based Phenotyping and Environmental Platforms

APPN Aerial SOP — Locked revision 1.0

Ground-Based Phenotyping and Environmental Platforms

In addition to UAV-based phenotyping platforms, APPN field operations make use of a range of **ground-based physiological, spectral, and environmental measurement tools** to support interpretation and validation of field phenomics data.

These tools provide complementary measurements that link remotely sensed observations with plant function and environmental conditions at the plot and plant scale.

Ground-based phenotyping tools

Ground-based phenotyping tools currently used, or commonly available, across APPN field nodes include:

- **Field spectroradiometers** for high-resolution reflectance and radiance measurements
- **Infrared gas analysers (IRGAs)** for measurement of photosynthesis, respiration, and gas exchange
- **Porometers** for stomatal conductance and transpiration assessment
- **Ceptometers** for canopy structure and light interception measurements
- **Chlorophyll meters and fluorometers** for assessment of photosynthetic status and stress responses
- **Field microscopes and imaging tools** for fine-scale visual inspection and trait validation

These instruments support targeted physiological and biochemical measurements that complement UAV-derived structural and spectral products.

Environmental monitoring

Environmental monitoring is an important component of field phenotyping, providing context for both ground-based and UAV observations.

Common instrumentation includes:

- **Automated weather stations** measuring variables such as air temperature, humidity, solar radiation, wind, and rainfall
- **Soil moisture probes** for monitoring soil water availability and temporal dynamics

These measurements help contextualise plant responses and support interpretation of phenotypic variation across time and sites.

Role within APPN field workflows

Ground-based phenotyping and environmental platforms are used to:

- Provide physiological context for UAV-derived traits
- Support validation and interpretation of spectral and structural products
- Enable deeper, mechanism-focused understanding of plant responses

- Assist comparison across sites, seasons, and experiments

These tools are often deployed in parallel with aerial phenotyping, but may also be used independently for focused measurement campaigns.

Standardisation and protocols (in development)

While many of these ground-based tools are routinely used across APPN field nodes, **formal, standardised protocols for their deployment, calibration, and reporting are still under development**.

Future protocol development will aim to:

- Define minimum metadata and reporting requirements
- Harmonise measurement approaches across nodes
- Support integration with UAV acquisition and processing workflows
- Improve comparability and reuse of physiological and validation measurements

Until standard protocols are published, users are encouraged to document methods and settings clearly at the project level and consult with relevant node staff.

Status of this page

> This page is intentionally maintained as a **placeholder** to acknowledge the role of ground-based phenotyping and environmental platforms within APPN field operations. Content will be expanded as standardised protocols and guidance documents are developed and released.